

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute `fork()`, understand `exec()` family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid < 0)
    {
        printf("Fork Failed!\n");
        return 1;
    }

    else if(pid == 0)
    {
        printf("\n--- Child Process ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
    }

    else
    {
        wait(NULL);

        printf("\n--- Parent Process ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }

    return 0;
}
```

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Before exec()\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("This line will not execute.\n");

    return 0;
}
```

```
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ ls
2520090074_Skill  destination.txt  practcal1.c  practical  practical2
OSPractical      os              practial1    practical1.c  practical2.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ mkdir OSSkill
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ cd OSSkill
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ nano skill1.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ gcc skill1.c -o skill1
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ ./skill1
```

--- Child Process ---

Child PID : 587

Parent PID : 586

--- Parent Process ---

Parent PID : 586

Child PID : 587

thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill\$ |

```
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ nano skill1b.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ gcc skill1b.c -o skill1b
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ ./skill1b
Before exec()
total 40
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16168 Aug 14 08:52 skill1
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 583 Aug 14 08:52 skill1.c
-rwxr-xr-x 1 thumu_hemanth_kumar thumu_hemanth_kumar 16000 Aug 14 08:54 skill1b
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 188 Aug 14 08:53 skill1b.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ |
```